DOW UNIVERSITY OF HEALTH SCIENCES

D.U.H.S

بسم الله الرحمن الرحيم

Resting Membrane Potential

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DIKIOHS ,DUHS.

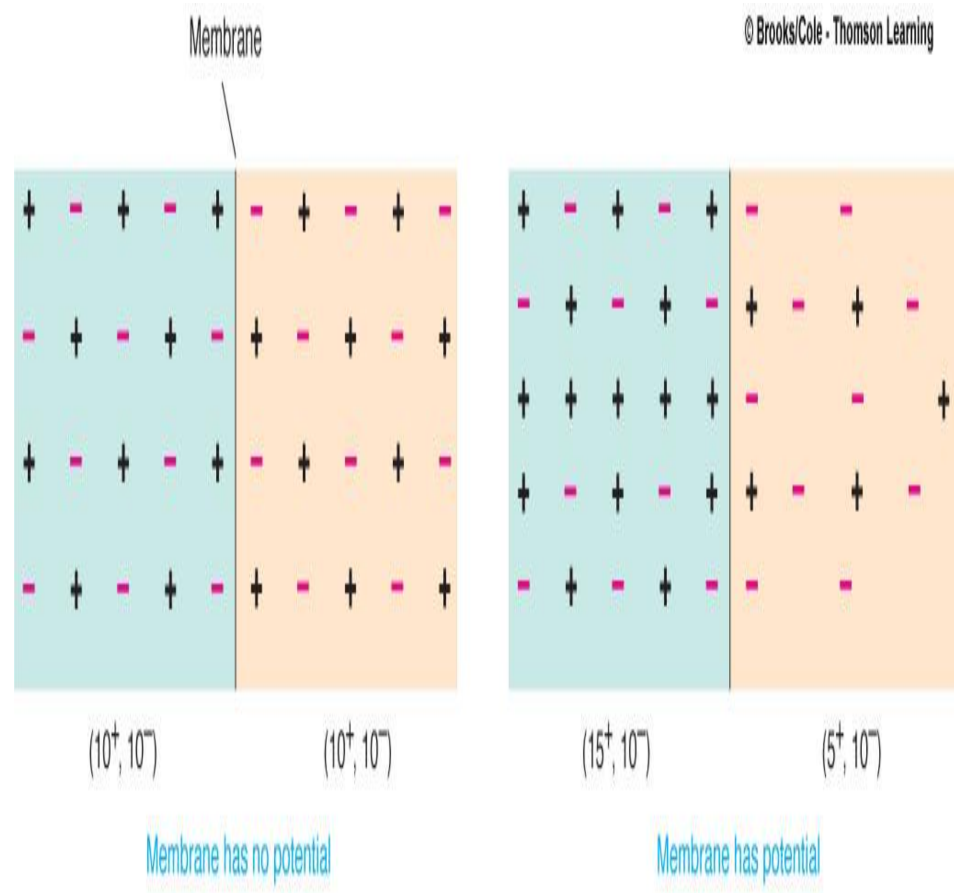
Learning outcomes

- Differentiate between membrane potential and resting membrane potential.
- Compare the basic types of ion channels, and explain how they relate to graded potentials and action potentials.
- Describe the factors that maintain a resting membrane potential.
- List the events that generate an action potential
- Explain how an action potential propagates along an axon
- Describe refractory period

Resting Membrane Potential

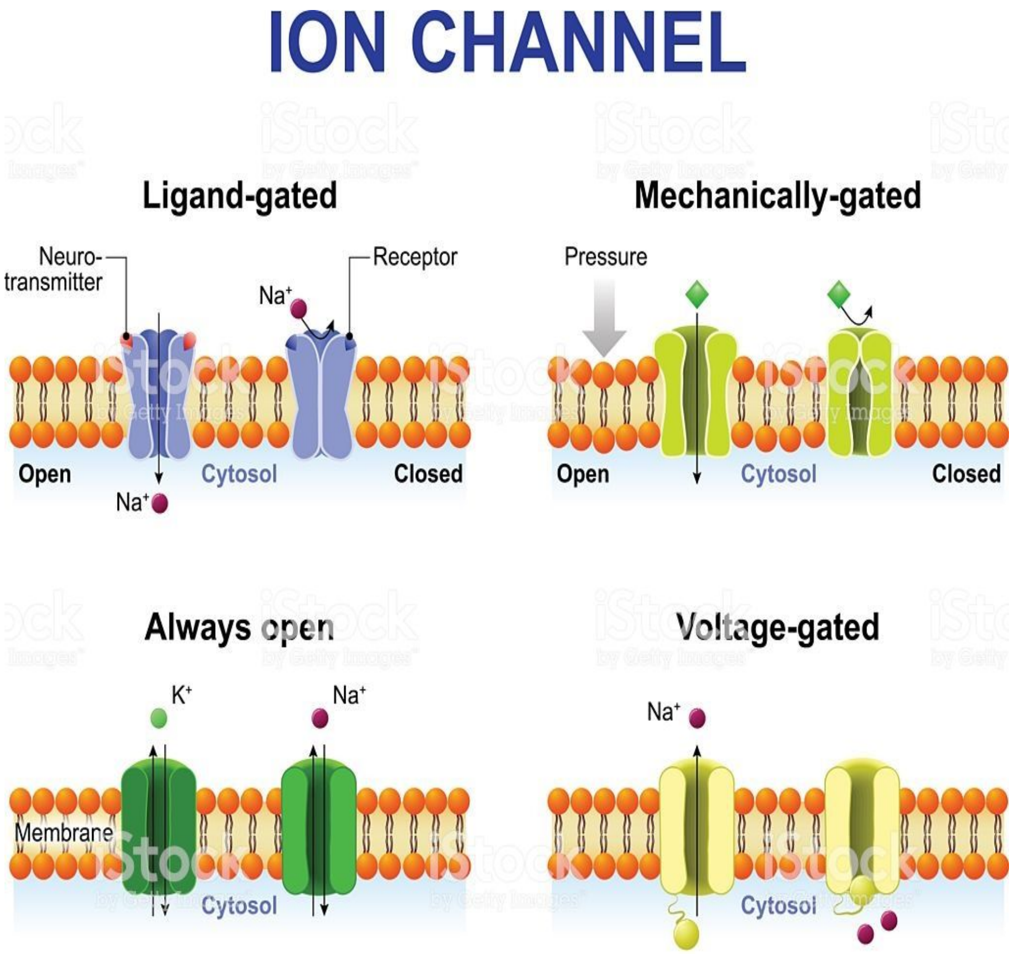
DEFINITION

- **Potential difference existing across the cell membrane under resting condition.**



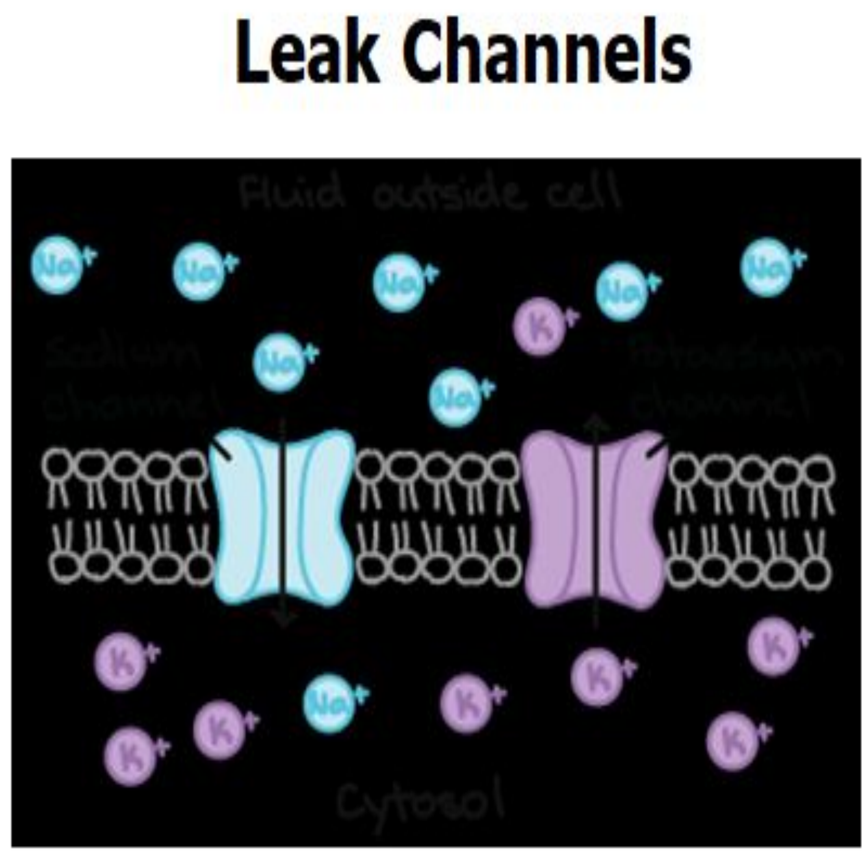
Types of Channels in the cell Membrane

- Leak Channels
- Voltage gated channels
- Ligand gated
- Mechanical gated(Hair Cells of inner Ear)



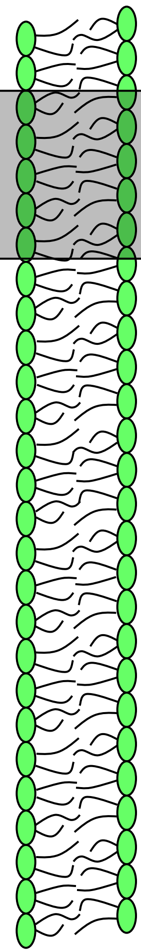
LEAK CHANNELS

- Sodium and potassium leak channels are responsible to establish RESTING MEMBRANE POTENTIAL



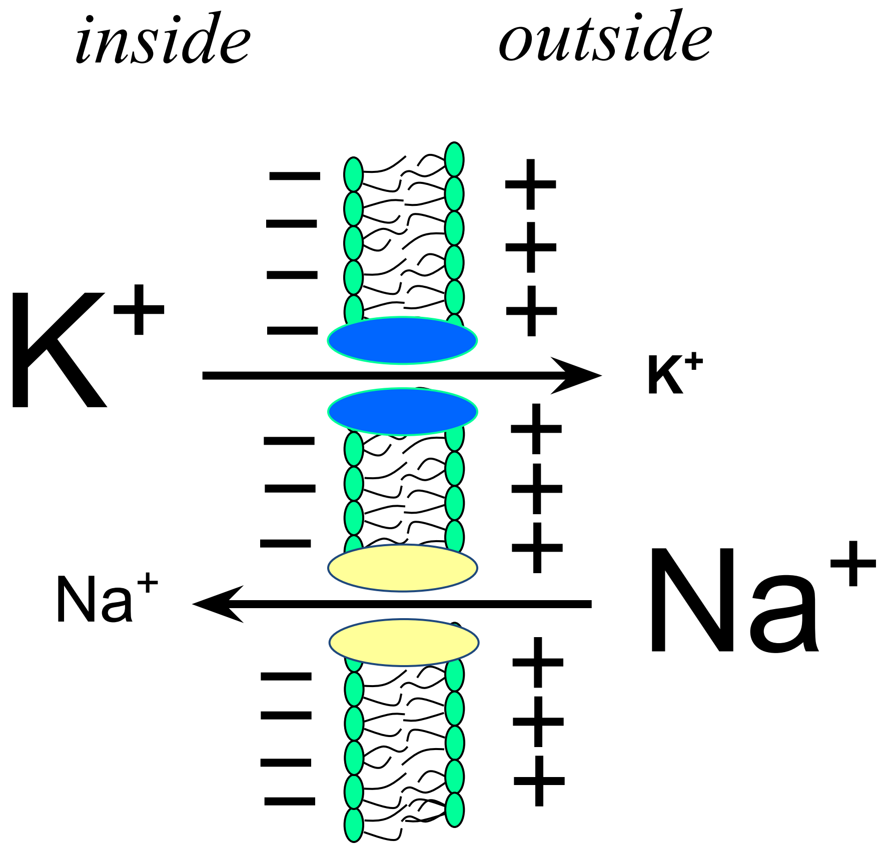
Molecular Gradients

	inside (in mM)	*outside* (in mM)
Na^+	14	142
K^+	140	4
Mg^{2+}	0.5	1-2
Ca^{2+}	10^{-4}	1-2
H^+	(pH 7.2)	(pH 7.4)
HCO_3^-	10	28
Cl^-	5-15	110
SO_4^{2-}	2	1
PO_3^-	75	4
protein	40	5



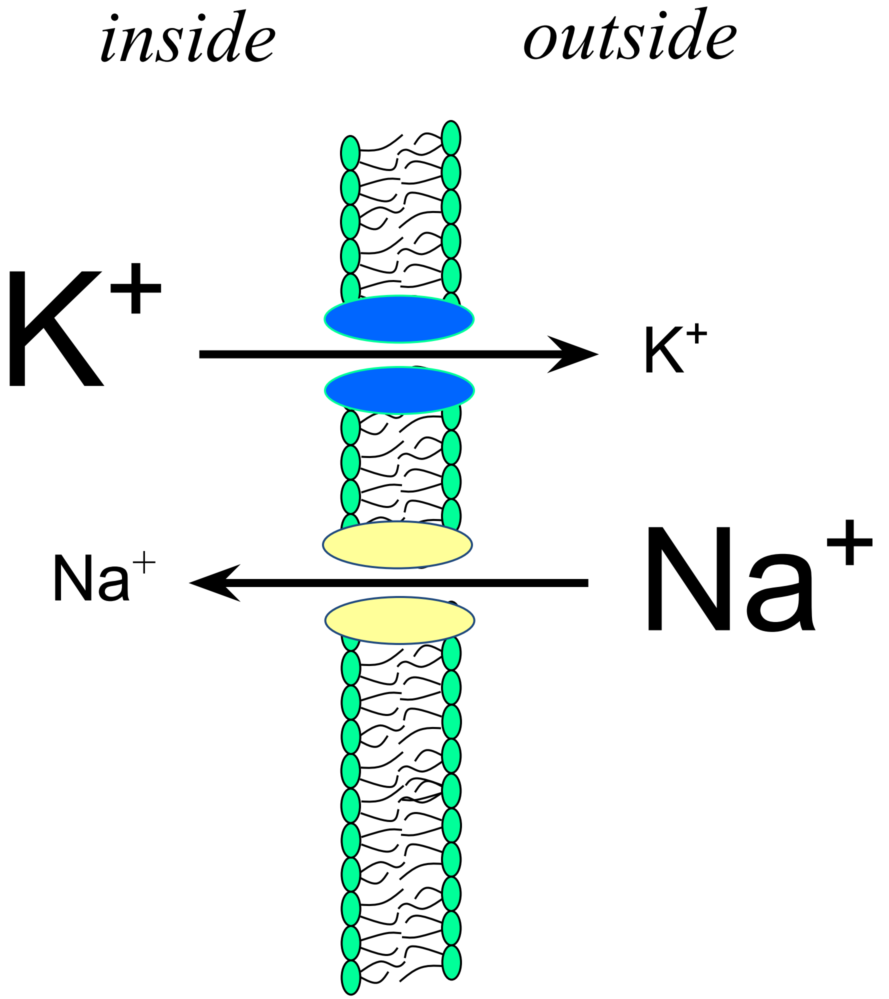
Membrane Potential (V_m):

- charge difference across the membrane -

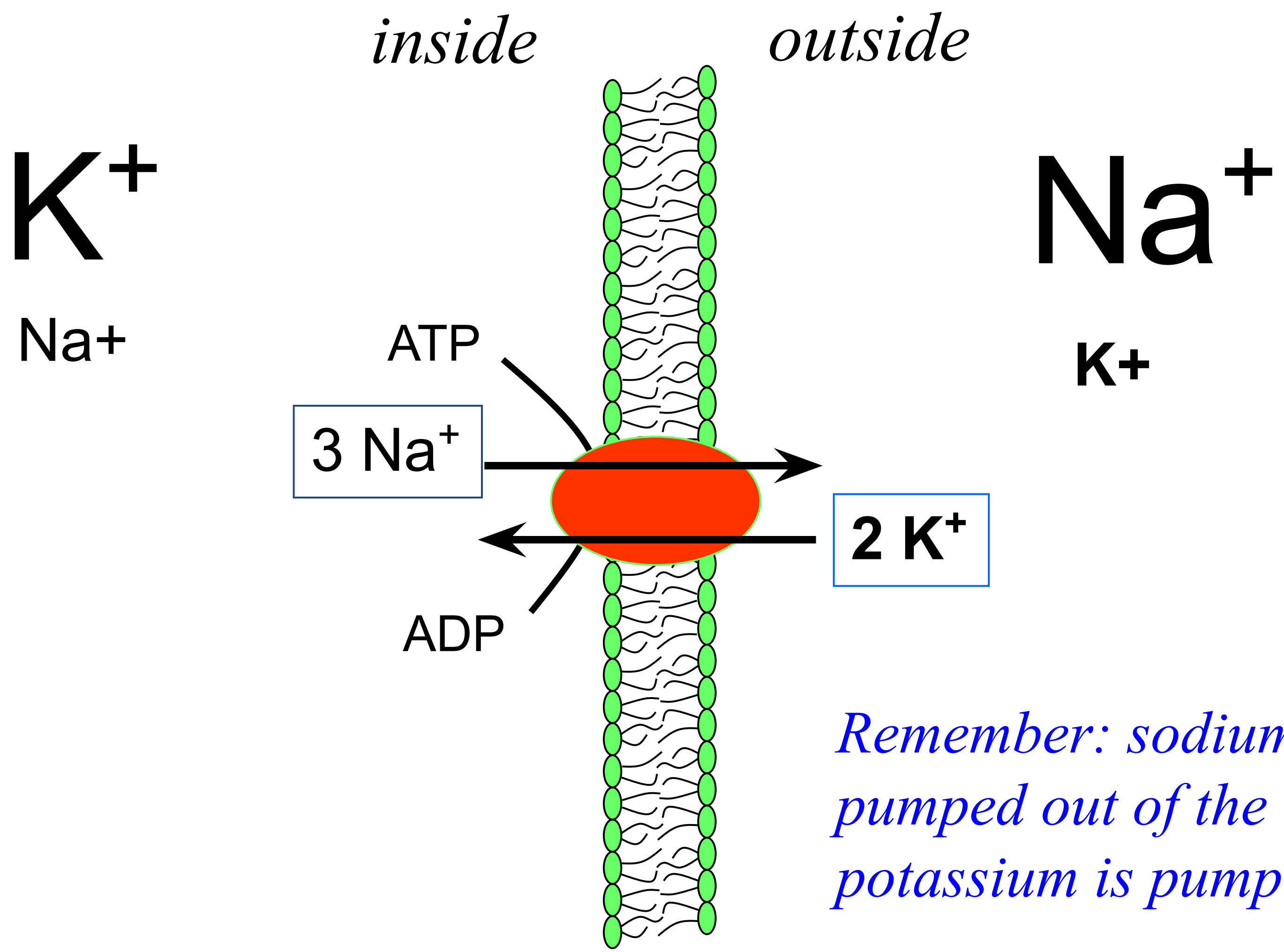


...how can passive diffusion of potassium and sodium lead to development of negative membrane potential?

Simple Diffusion

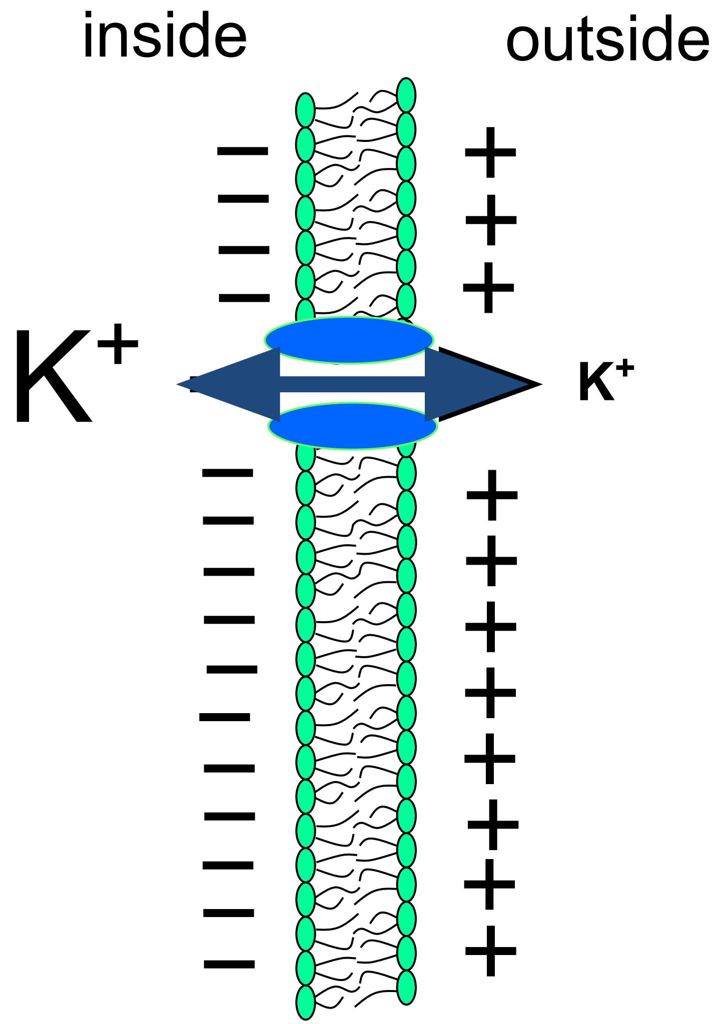


Active Transport



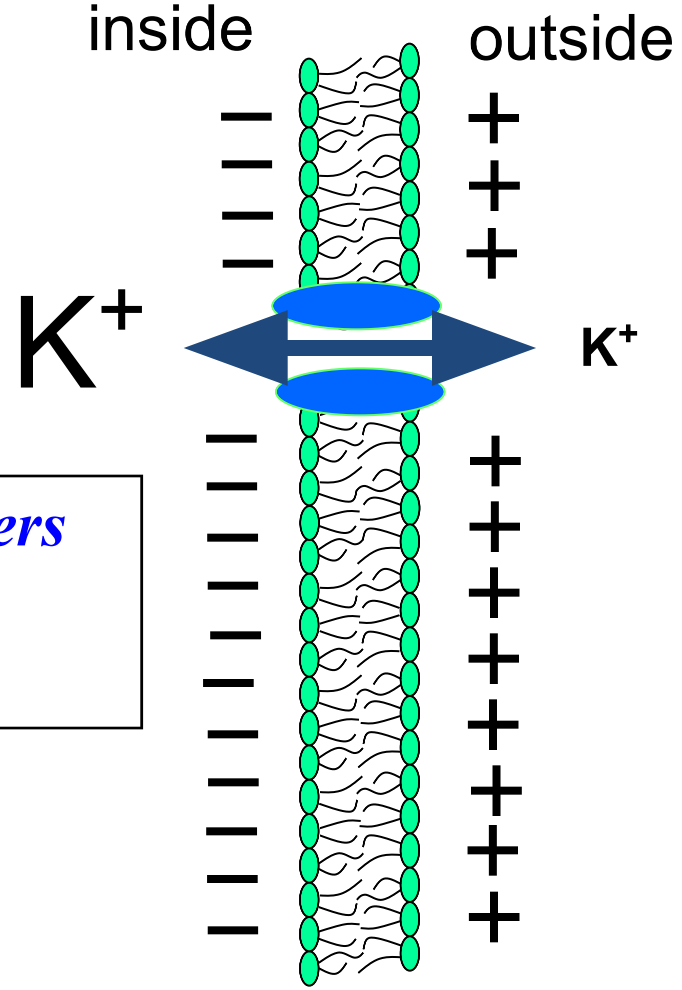
Remember: sodium is pumped out of the cell, potassium is pumped in...

If a membrane were permeable to only K^+ then…



K^+ would diffuse down its concentration gradient until the electrical potential across the membrane countered diffusion.

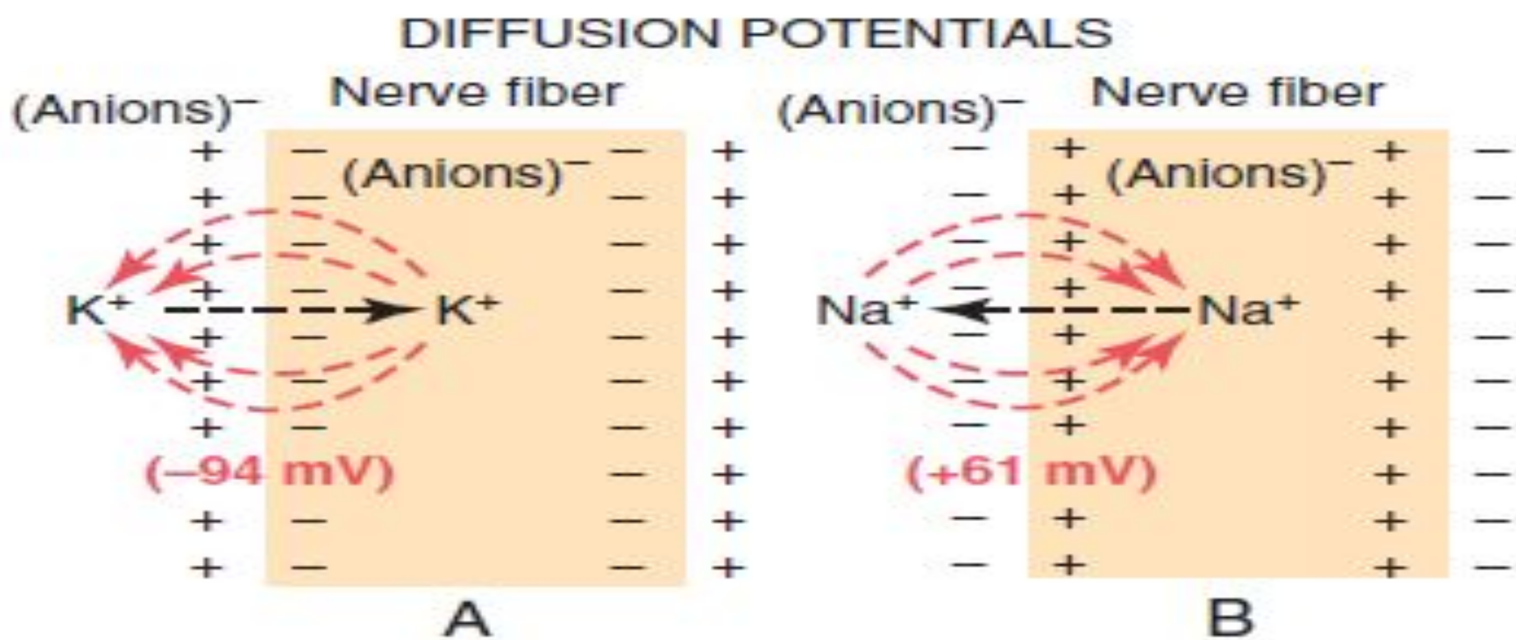
If a membrane were permeable to only K^+ then…



The electrical potential that counters net diffusion of K^+ is called the K^+ equilibrium potential (E_K).

MEMBRANE POTENTIALS CAUSED BY ION CONCENTRATION AND ELECTRICAL GRADIENT

Differences Across a Selectively Permeable Membrane



igure 5-1. A, Establishment of a diffusion potential across a nerve iber membrane, caused by diffusion of potassium ions from inside he cell to outside through a membrane that is selectively permeable

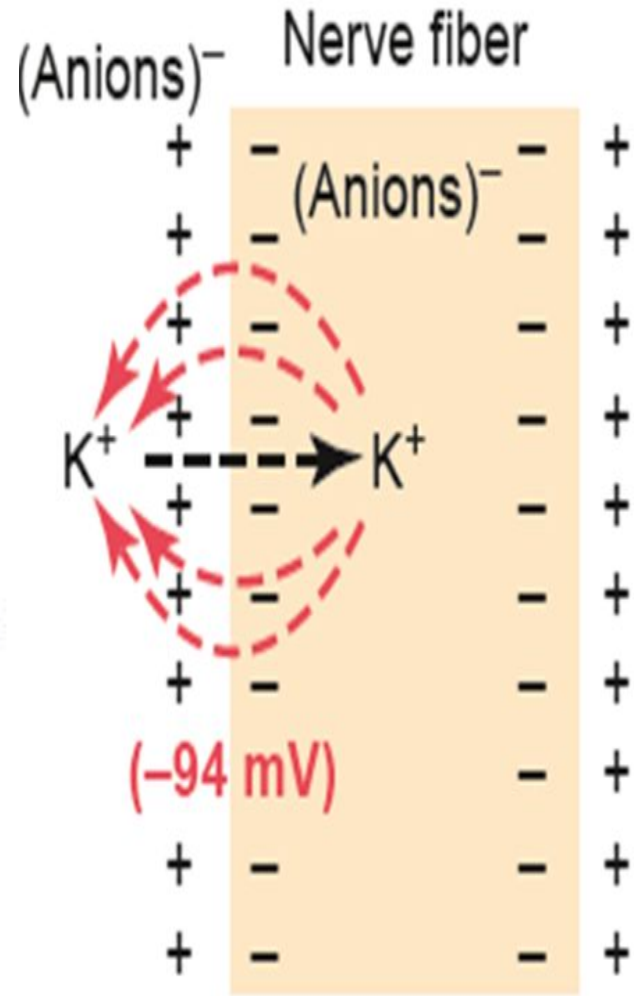
The Nernst Equation

NERNST POTENTIAL/ EQUILIBRIUM POTENTIAL

- The diffusion potential across a membrane that exactly opposes the net diffusion of a particular ion called the Nernst potential for that ion
- **NERNST EQUATION**: Used to calculate the equilibrium potential of single ion

$$\text{EMF (millivolts)} = \pm \frac{61}{z} \times \log \frac{\text{Concentration inside}}{\text{Concentration outside}}$$

where *EMF* is electromotive force and *z* is the electrical charge of the ion (e.g., +1 for K^+).



The Potassium Nernst Potential

...also called the equilibrium potential

$$E_K = -61 \log \frac{K_i}{K_o}$$

Example: If K_o = 4 mM and K_i = 140 mM

E_K = -61 log(140/4)

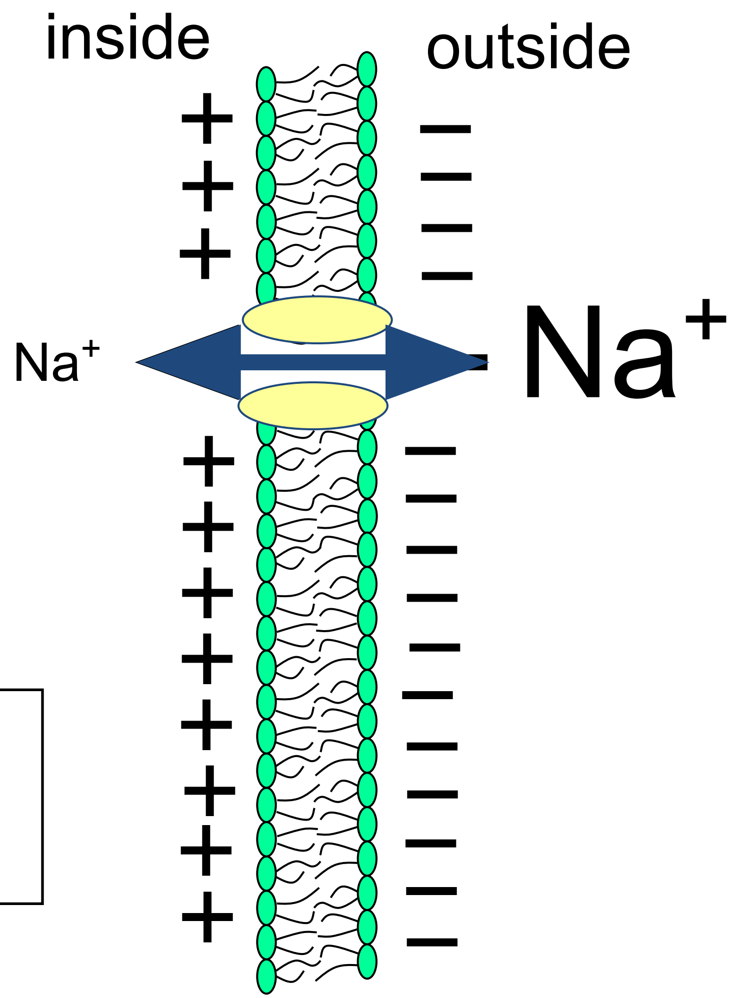
E_K = -61 log(35)

E_K = -94 mV

So, if the membrane were permeable only to K+, Vm would be -94 mV

If a membrane were permeable to only Na^+ then…

Na^+ would diffuse down its concentration gradient until potential across the membrane countered diffusion.



The electrical potential that counters net diffusion of Na^+ is called the Na^+ equilibrium potential (E_{Na}).

The Sodium Nernst Potential

$$E_K = -61 \log \frac{Na_i}{Na_o}$$

Example: If Na_o = 142 mM and Na_i = 14 mM

E_K = -61 log(14/142)

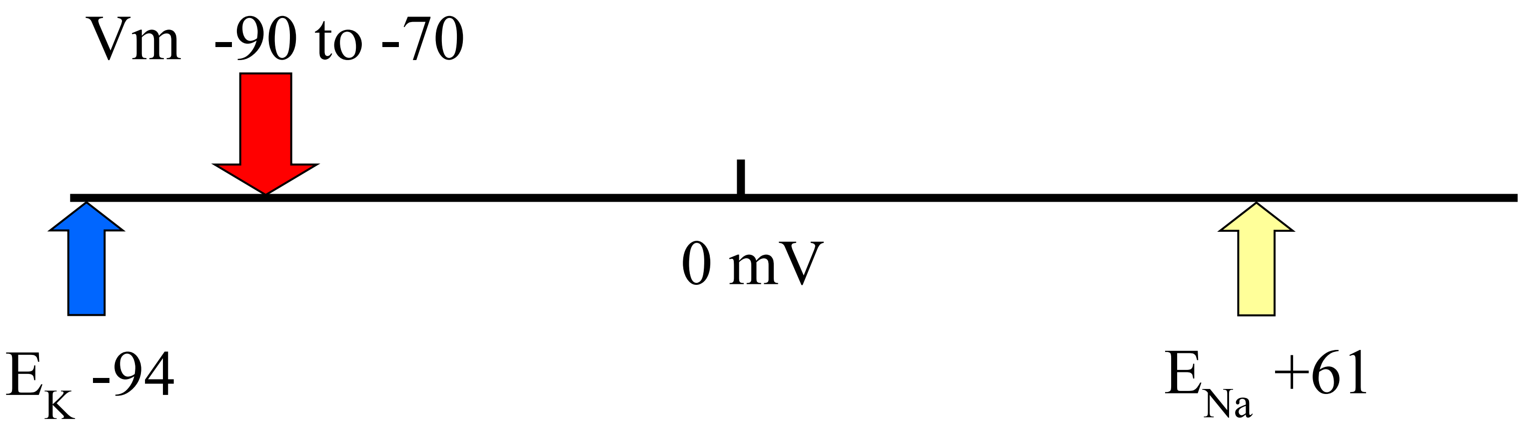
E_K = -61 log(0.1) =(-1)

-61x-1=+61

E_K = +61 mV

So, if the membrane were permeable only to Na^+,
Vm would be +61 mV

Resting Membrane Potential



Why is Vm so close to E_K?
Ans. The membrane is far more permeable to K than Na..

Goldman equation:

- **Used to calculate the equilibrium potential of 2 or more ions**

$$\text{EMF} = -61 \cdot \log \frac{p'_K [K^+]_i + p'_{Na} [Na^+]_i + p'_{Cl} [Cl^-]_o}{p'_K [K^+]_o + p'_{Na} [Na^+]_o + p'_{Cl} [Cl^-]_i}$$

- In summary, the diffusion potentials caused by potassium and sodium diffusion would give a membrane potential of about −86 millivolts

The Goldman-Hodgkin-Katz Equation

(also called the Goldman Field Equation)

Calculates V_m when more than one ion is involved.

$$V_m = 61 \cdot \mathbf{log} \frac{p'_K [K^+]_o + p'_{Na} [Na^+]_o + p'_{Cl} [Cl^-]_i}{p'_K [K^+]_i + p'_{Na} [Na^+]_i + p'_{Cl} [Cl^-]_o}$$

or

$$V_m = -61 \cdot \mathbf{log} \frac{p'_K [K^+]_i + p'_{Na} [Na^+]_i + p'_{Cl} [Cl^-]_o}{p'_K [K^+]_o + p'_{Na} [Na^+]_o + p'_{Cl} [Cl^-]_i}$$

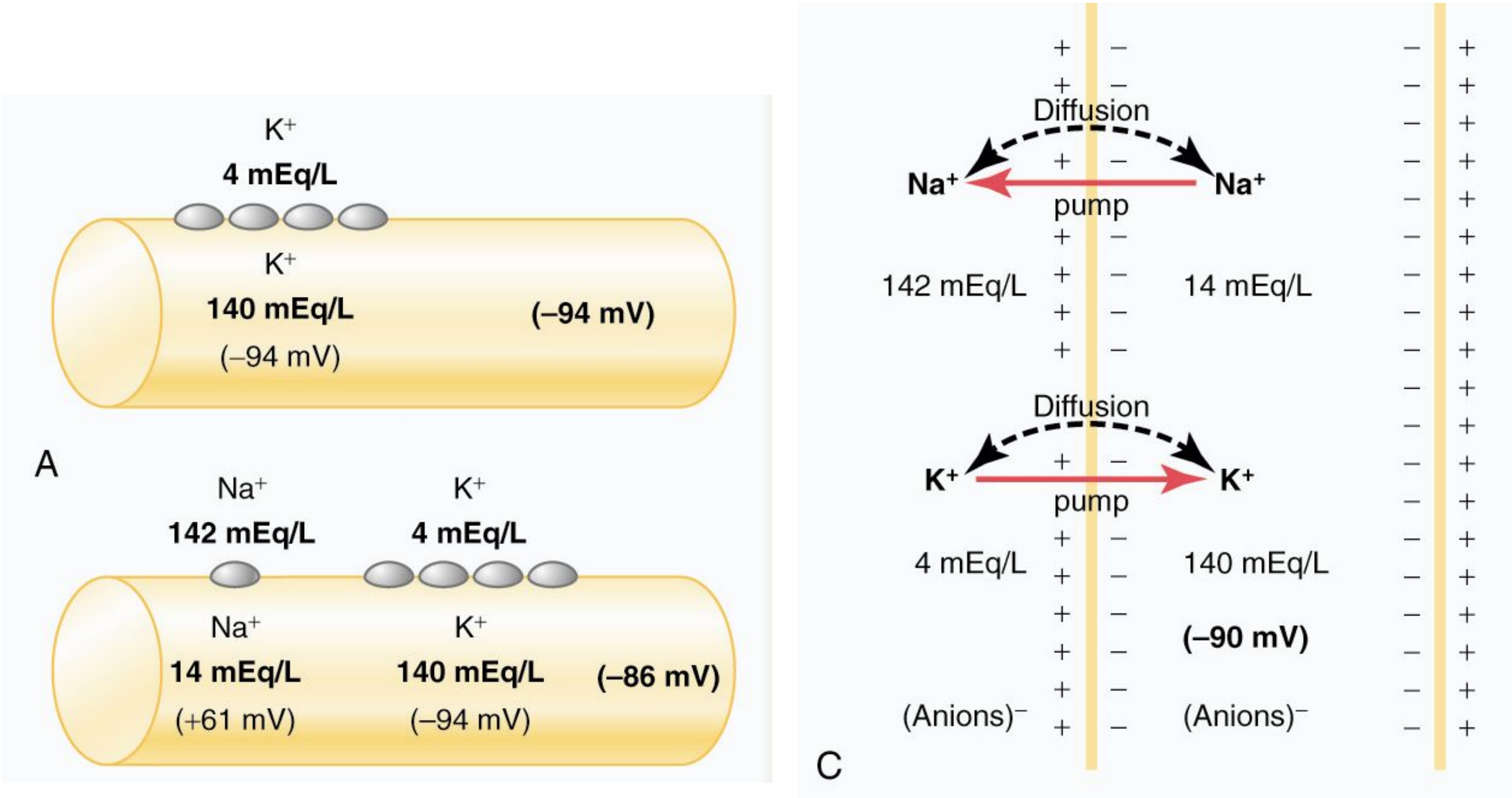
NOTE:

P' = permeability

The Goldman-Hodgkin-Katz Equation

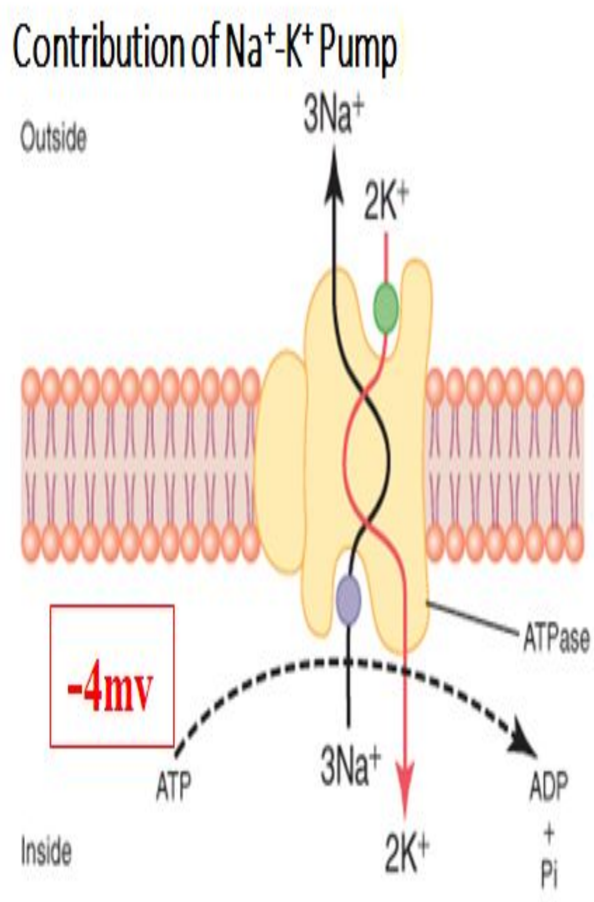
The resting membrane potential is closest to the equilibrium potential for the ion with the highest permeability.

Resting Membrane Potential Summary



Functions of Na+ - K+ Pump (ATPase)

- 1. As an Enzyme (Cleaves ATP to ADP)
- 2. Electrogenic. (Contributes -4mv in RMP)
- 3. Homeostasis of Main Electrolytes and water & hence maintains volume of the cell.



Summary of RMP calculation

- Nernst potential for Potassium -94mv
- Nernst potential for Sodium +61mv
- Putting these values in Gold man equation, gives a value of -86mv

Which is nearer to K+ diffusing potential

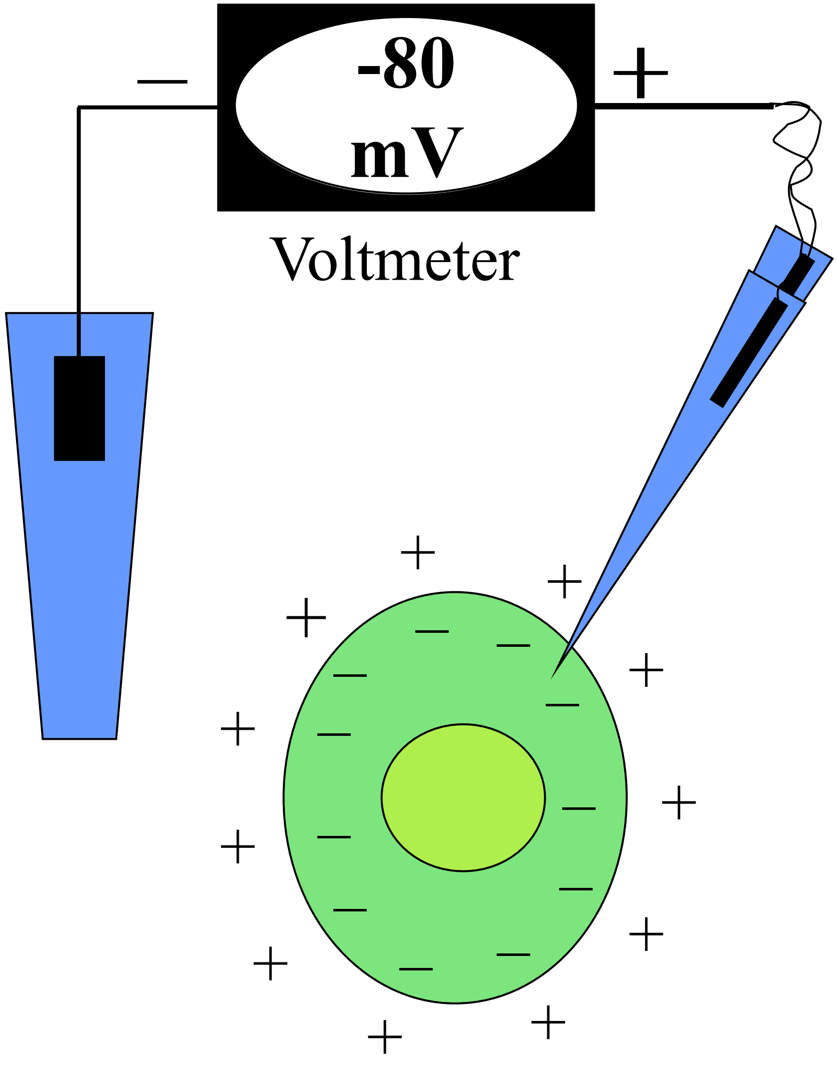
- Na- K pump provides - 4mv
- i.e adding -86 and -4mv= -90mv
- Resting membrane potential in nerves is -90 mv

KEY POINTS

- Sodium and potassium leak channels are responsible to establish RESTING MEMBRANE POTENTIAL(RMP).
- Na+ concentration outside cell is 142meq/l
- K+ concentration inside cell is 140 meq/l
- Nearnst potential for sodium is +61
- Nearnst potential for potassium is -94
- To calculate equilibrium potential for a single ion Nearnst Equation is used
- To calculate equilibrium potential for 2 or more ions Goldmann's equation is used
- Resting membrane potential of nerves is -90mv
- Na-K pump contributes -4mv to establish RMP(out of -90).

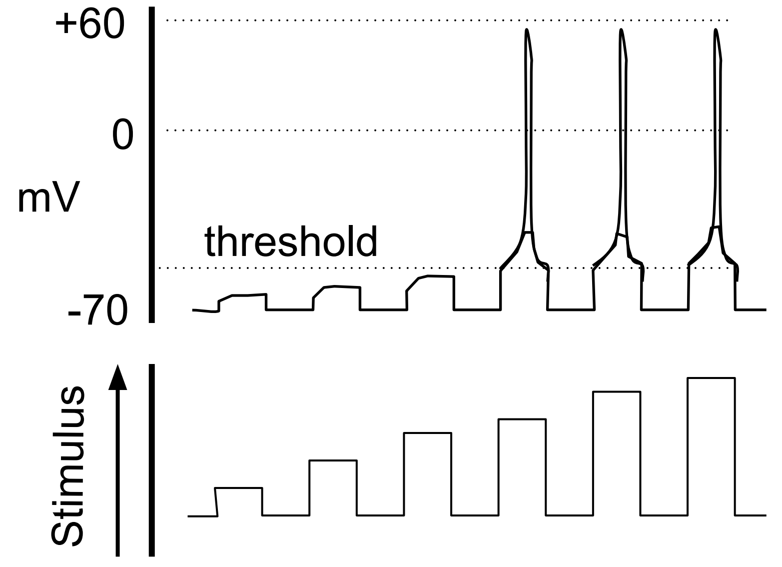
Resting and action potentials

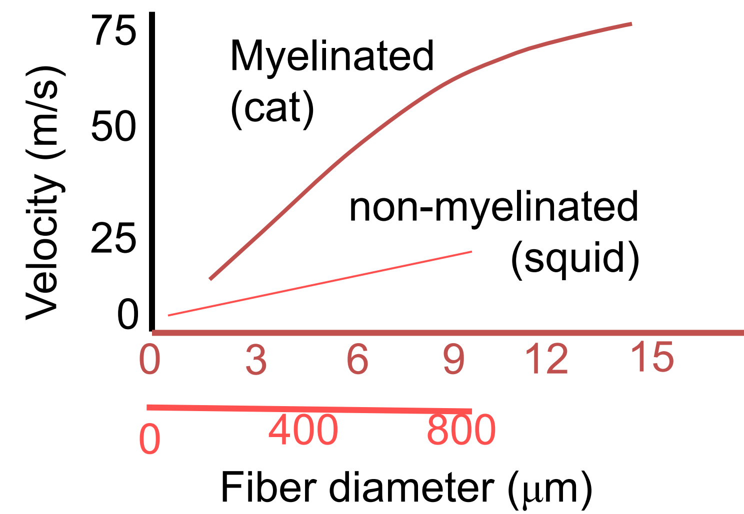
- Recall that cells:
  - contain high K^+ concentration
  - have membranes that are essentially permeable to K^+ at rest
- Membrane electrical potential difference *(membrane potential)* is generated by diffusion of K^+ ions and charge separation
  - measured in mV (=1/1000th of 1V)
  - typically resting membrane potentials in neurons are -70 to 90 mV



Properties of action potentials

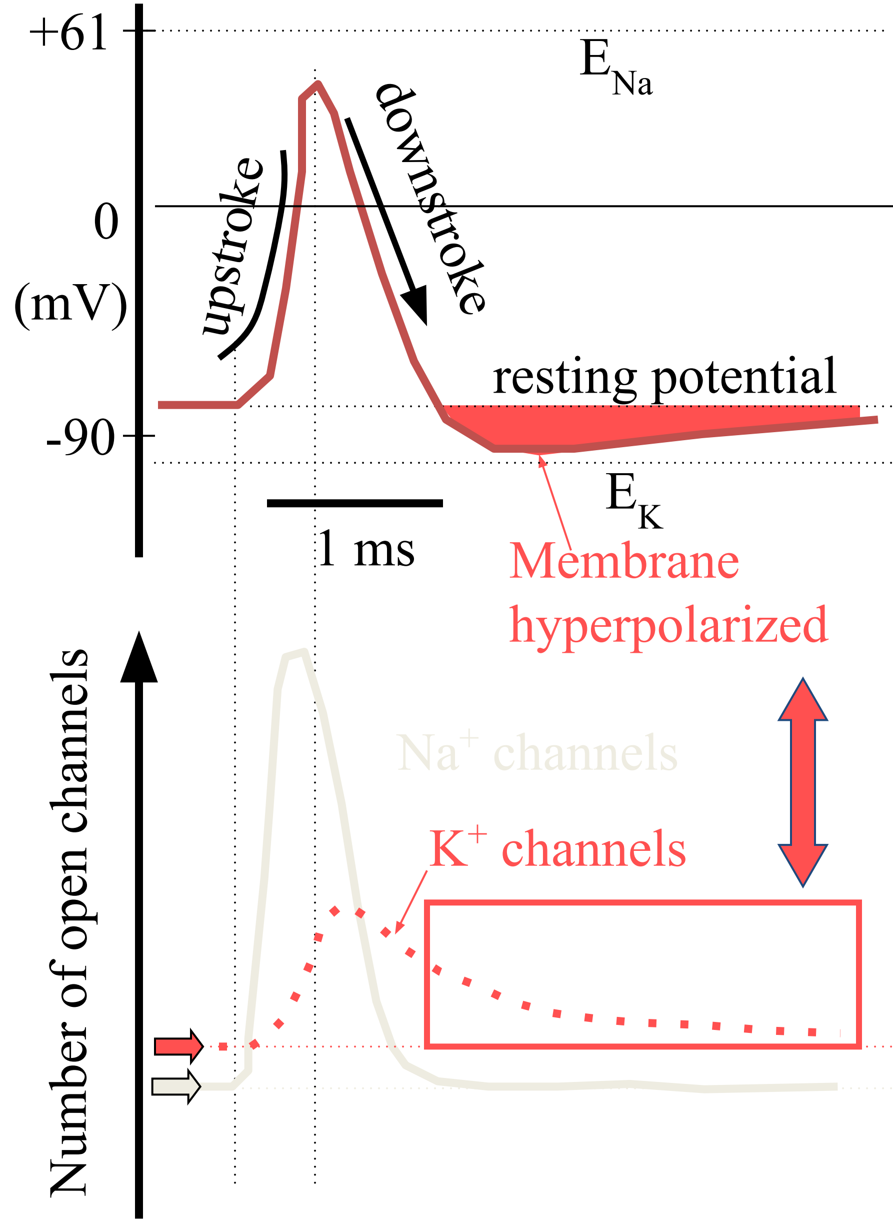
- Action potentials:
  - are all-or-none events
    - threshold voltage (usually 15 mV positive to resting potential)
  - are initiated by depolarization
    - action potentials can be induced in nerve and muscle by extrinsic (percutaneous) stimulation
  - have constant amplitude
    - APs do not summate - information is coded by frequency not amplitude.
  - have constant conduction velocity
    - True for given fiber. Fibers with large diameter conduct faster than small fibers. As a general rule:
      - **myelinated** fiber diameter (in mm) x 4.5 = velocity in m/s.
      - Square root of **unmyelinated** fiber diameter = velocity in m/s





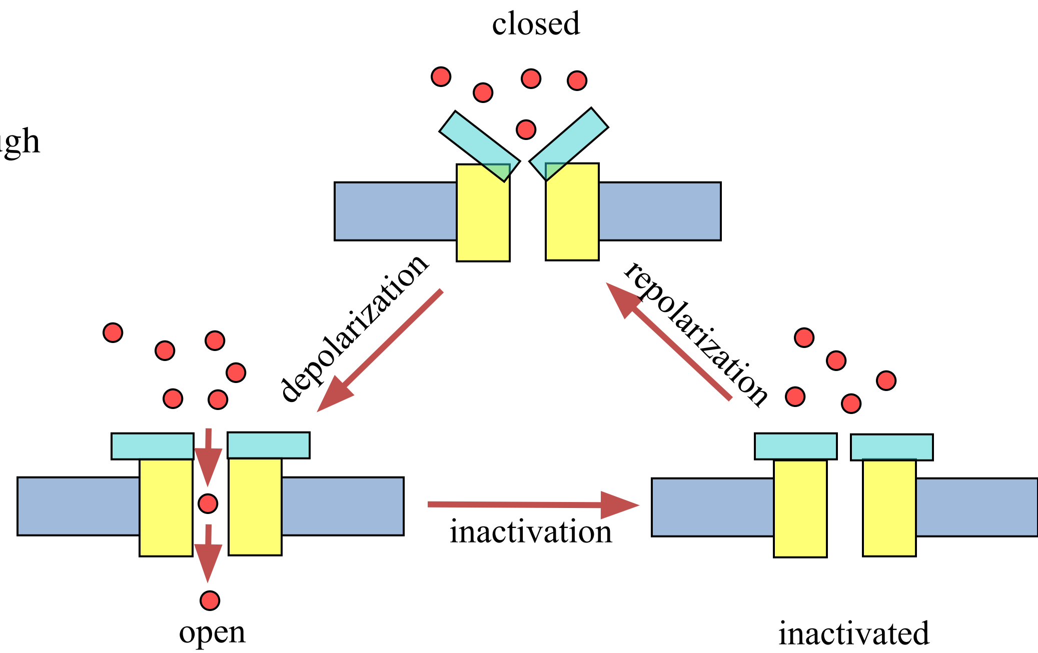
The AP - membrane permeability

- During the upstroke of an action potential:
  - ❑ Na permeability increases
    - due to opening of Na^+ channels
      - memb. potential approaches E_{Na}
- During the downstroke of an action potential:
  - ❑ Na permeability decreases
    - due to inactivation of Na^+ channels
  - ❑ K permeability increases
    - due to opening of K^+ channels
      - mem. potential approaches E_K
- After hyperpolarization of membrane following an action potential:
  - ❑ not always seen!
  - ❑ There is increased K^+ conductance
    - due to delayed closure of K^+ channels



Ion channels

- Ion channels - structure
  - proteins that span the membrane
  - have water filled channel that runs through protein
- Ion channel - properties
  - Have conducting states and non-conducting states
  - transition between states = ‘gating’



- channels ‘gate’ in response to:
  - changes in membrane potential (usually depolarization)
    - voltage-gated channels. Action potential propagation relies on voltage-gated channels
  - occupation of receptor
    - ligand-gated or receptor operated channels (ROCs). These initiate action potentials
  - mechanical forces
    - mechanosensitive channels - important for hearing.

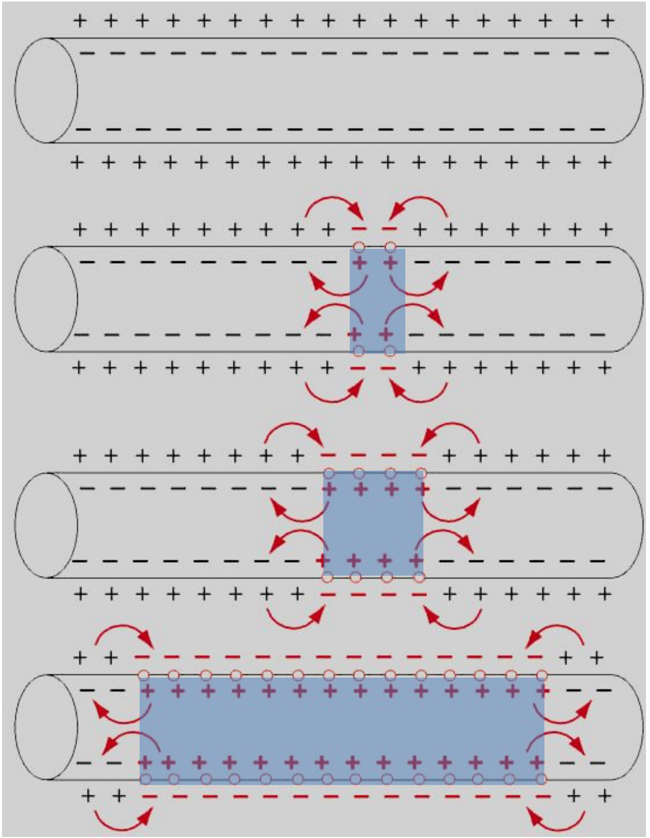
Propagation:

Opening of Na^+ channels generates local current circuit that depolarizes adjacent membrane, opening more Na^+ channels…

Rest

Stimulated
(local depolarization)

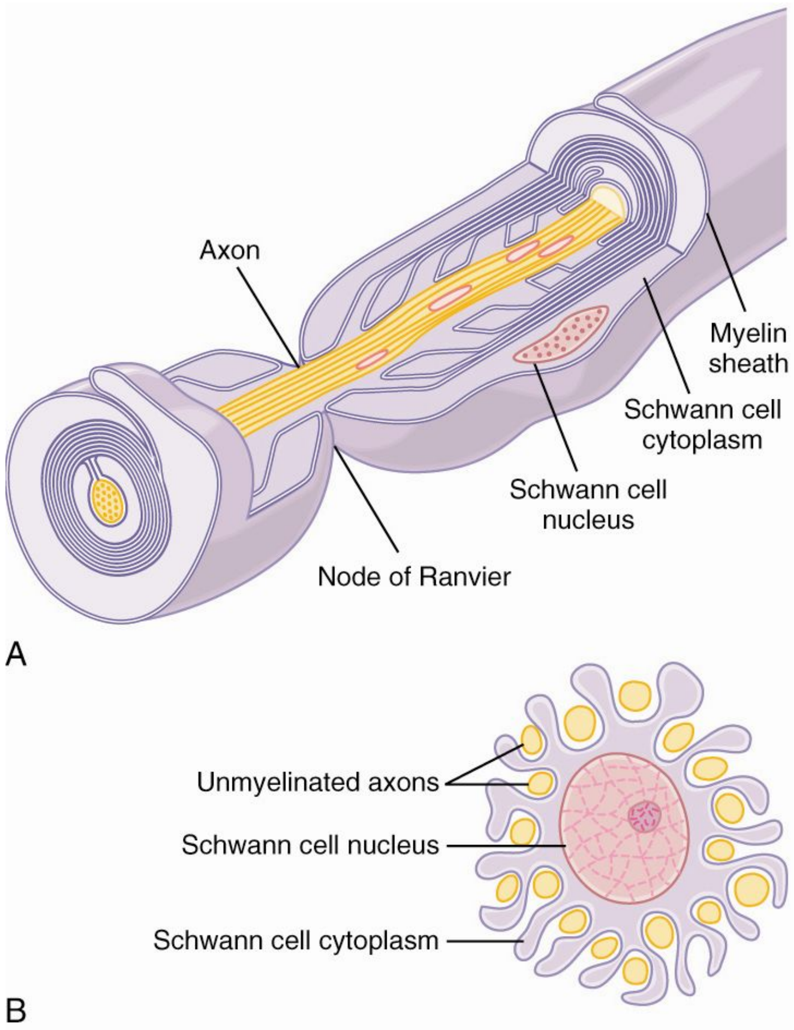
Propagation
(current spread)



Signal Transmission:

Myelination

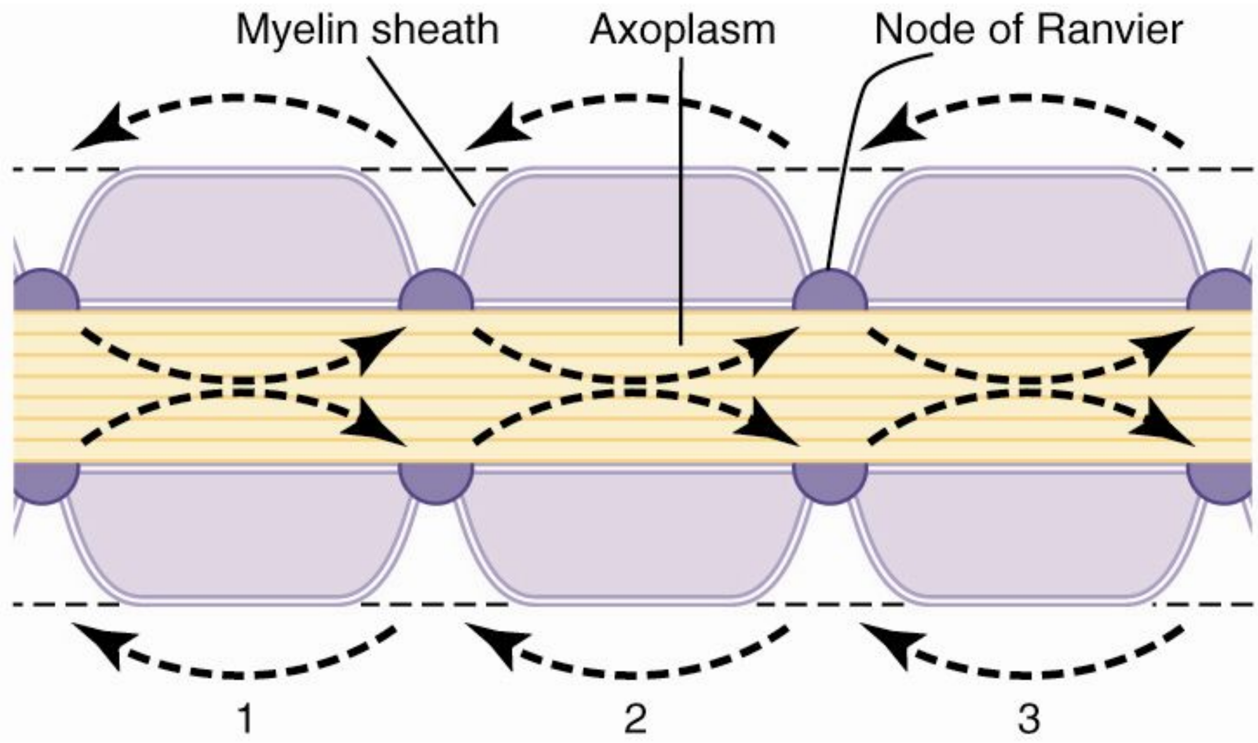
- **Schwann cells** surround the nerve axon forming a myelin sheath
- Sphingomyelin decreases membrane capacitance and ion flow 5,000-fold
- Sheath is interrupted every 1-3 mm : **node of Ranvier**



- At intervals of 1 to 2 mm, there are breaks in the myelin sheath, at the nodes of Ranvier. At the nodes, membrane resistance is low, current can flow across the membrane, and action potentials can occur. Thus, conduction of action potentials is faster in myelinated nerves than in unmyelinated nerves because action potentials “jump” long distances from one node to the next, a process called Saltatory Conduction.

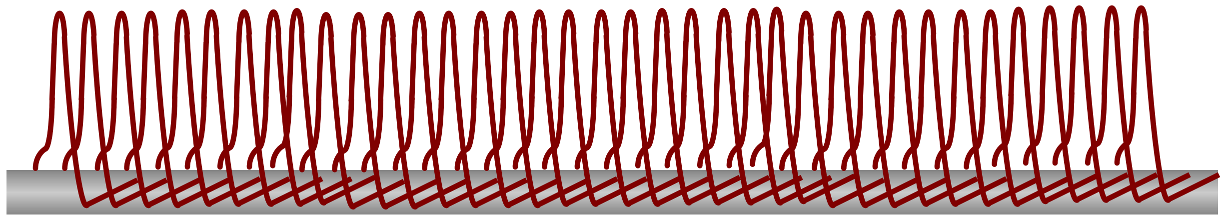
Saltatory Conduction

- AP's only occur at the nodes *(Na channels concentrated here!)*
- increased velocity
- energy conservation

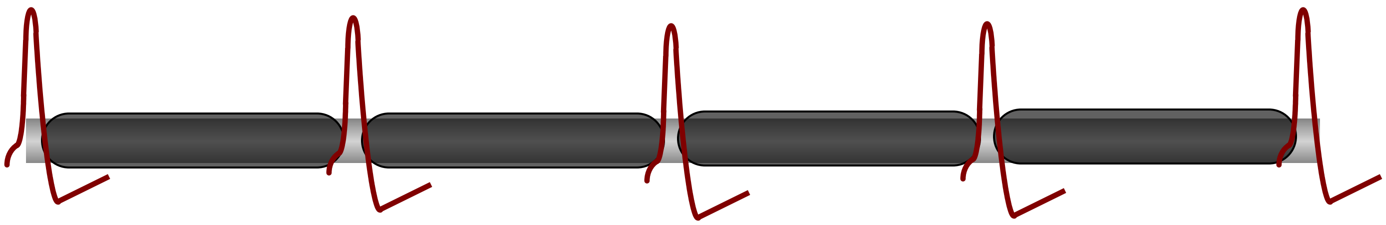


Conduction velocity

- non-myelinated vs myelinated -

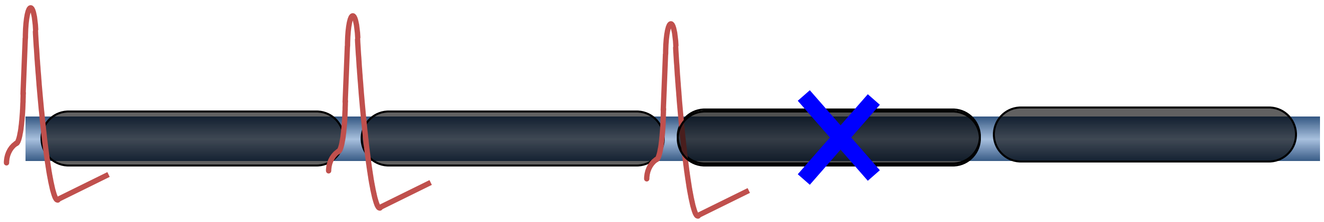


non-myelinated



myelinated

Multiple Sclerosis



- *MS is an immune-mediated inflammatory demyelinating disease of the CNS -*

- *About 1 person per 1000 in US is thought to have the disease - The female-to-male ratio is 2:1 - whites of northern European descent have the highest incidence*

Patients have a difficult time describing their symptoms. Patients may present with paresthesias of a hand that resolves, followed in a couple of months by weakness in a leg or visual disturbances. Patients frequently do not bring these complaints to their doctors because they resolve. Eventually, the resolution of the neurologic deficits is incomplete or their occurrence is too frequent, and the diagnostic dilemma begins.

References:

Guyton & Hall, Text Book of Medical Physiology 12th Edition.

Ganong's review of Medical Physiology 24th Edition

THANK YOU